

Meeting Summaries of Jan 29

Date: January 29

Time: 14:30 - 15:00 (PDT)

Attendees: Prof. Jan, Dr. Javed, Dr. Yue and Feng

Topic: Discussion about the reproduction work on Jian's study results, the **BE**nthic **A**ssessment of **S**edimen**T** (the BEAST) with multivariate approach for the **SCDRS** (St. Clair-Detroit River System) sediment contamination assessment.

Summary of Discussion

The reproduced results that were discussed in this meeting are **SCDRS case only**.

The Detroit River case will be reproduction work, which would follow the same BEAST approach by dropping off the non-DR sites and including the 'true' velocity values.

0.1 Some unclear points to clarify.

- Prof. Jan asked reason about the **three clusters** chosen from hierarchical clustering over the SCDRS case that was from Feng's previous document.

It is unclear because in hierarchical clustering on these ref-sites, **1)** Jian selected bifurcation for the SCDRS case, **2)** later Jian chose 3 clusters for the Detroit River case¹, not for the SCDRS case.

- Prof. Jan and Feng compared the **two reproduced** Ward's clustering with the **original** Jian's clustering results.

The two reproduced Ward's clustering results are with Euclidean and Manhattan, Jian used Manhattan (cityblock) distance for her clustering. In reproduction work, **however**, only the Euclidean setting gives the exact same clustering result as Jian's, while the Manhattan setting gives different but similar clustering (e.g., site '47FB' went to different groups).

0.2 The good reproduced results to remain

- Prof. Jan suggested to **keep the same clustering result** as Jian's original one.

In the SCDRS case, they are the 62 ref-sites bifurcated into 2 clusters (55 vs 7) that are partially informed by the bad classification between two specific groups when choosing 3 clusters.

In the Detroit River case, Jian chose 3 clusters for the ref-sites.

¹The Detroit River case will Feng's next reproduction work, which is almost same as the BEAST of multivariate approach for the SCDRS case.

- The reproduced **Taxa ANOVA table** across the 2 clusters are acceptably consistent with Jian's original one.
- The reproduced **DFA model** with 5 environmental variables for 2 clusters is acceptably consistent with Jian's original one.
- The **confusion matrix** of the DFA on the 62 ref-sites is acceptably consistent with Jian's original one.
- **RDA triplot** is an extension work in revealing taxa-environment relationships. This RDA triplot makes a good sense and clear interpretation to elucidate the (continuous) relationships between each taxa abundance and environmental features.

In SCDRS case, the RDA was built on the 62 ref-sites with around **10 sites** from Detroit River that have raw velocity values, **the rest sites have imputed velocity values**.

In Detroit River case, this imputed velocity issue will be fixed because all DR sites have raw velocity values.

0.3 Improvements and extensions to work on

- In the **hierarchical clustering** work, it is worth to assess the robustness of the clustering results and refine it with the **PV clustering method**.
- In the zoobenthic community index work, it needs to align with Jian's **Bray-Curties Ordination method with NMDS**.

NMDS is a non-parametric method that preserves the rank order of dissimilarities. It does not require specific geometry dissimilarity measures (like Euclidean or other Metrics) to arrange the sites in a low-dimensional taxa space meanwhile preserving their multivariate (how-dimensional) relationships.

Action Items

- To the clustering in SCDRS case, Feng will **remain the choice of bifurcation** (55 vs 7) and **clarify his computation way** of reproducing the same result (is it "Euclidean" or "Manhattan" distance in the reproduction?) as Jian's original one.
- In the same clustering work, Feng will implement the **PV clustering method** to assess the robustness of the clustering results and consider how to refine clustering with it.
- From the RDA triplot work, Feng will derive the ecological interpretations to elucidate the relationships between taxa abundance and environmental features, which also helps validate the DFA model results.
- In the zoobenthic community index work, Feng will **reproduce the Bray-Curties Ordination method with NMDS** to align with Jian's original work.
- **To the Detroit River case**, Feng will reproduce the same BEAST of multivariate approach with the almost same data structure as the SCDRS case, but leveraging the known **raw velocity values** for all DR sites.

Follow-Up

Temporary meeting time: February 5, 14:30 - 15:00 (PDT)

Supportive materials

- A meeting recording with subtitles has been restored on Feng's local laptop.